PROJECT REPORT

MOSQUITO REPELLENT SYSTEM BASED ON POWER ELECTRONICS

TAHSIN JAWAD KHAN-200021207
K M ISTIAQUE-200021220
MD FIROZ HOSSAIN-200021225
SHAFAYET SADIK SOWAD-200021248

Objective:
To create a zapper which would electrify a net.

Introduction:
In this project which has been assigned to us, we are required to implement our knowledge of power electronics and come up with solutions for real-world problems. The inspiration for this project came to us one fine evening when the net of our dorm room was torn and we had no mosquito bats , so we had no chance but to just sit and suffer through being bitten by millions of mosquitos during that entire time.

So, moving forth, we engineered a zapper which could be attached to the net and then we can attach it to the windows ,it will have high frequency repellent system so mosquitos can not come near the window.Even if they come near and try to pass through it shall not pass because it will get electrocuted on the high voltage net. It can be used anywhere from homes, to schools , to hospitals and dorm rooms and the best part of our new **electric mosquito repellent net** is that the circuit of this net can be easily built and it is well documented here, which would allow everybody to build this net.

Theoretical Background:
The theoretical background of the project involves a comprehensive understanding of the key components used in the development of the electric mosquito repellent net. Each component plays a crucial role in creating an effective and efficient system for repelling mosquitoes.

1. 555 Timer:
The 555 timer is a versatile integrated circuit widely used in electronic projects for generating accurate time delays and pulses. In the context of the mosquito repellent net, the 555 timer is employed to generate a high-frequency pulse signal. This high-frequency signal is essential for creating a repellent system that is effective in deterring mosquitoes from approaching the window.

2. Bridge Rectifier:

The bridge rectifier is utilized to convert alternating current (AC) into direct current (DC). In this project, the bridge rectifier ensures a unidirectional flow of current, providing a stable DC power source for the subsequent components. This is crucial for the reliability and stability of the mosquito repellent net.

3. Transformer:

The transformer is an integral part of the project, serving to step up or step down the voltage as required. It plays a vital role in adjusting the voltage levels to suit the specific needs of the circuit. The transformer facilitates the conversion of power and ensures that the system operates efficiently and safely.

4. Mosfet Switch:

The Metal-Oxide-Semiconductor Field-Effect Transistor (MOSFET) switch acts as a crucial component in the circuit. It functions as an electronic switch that controls the flow of current. In this project, the MOSFET switch is used to manage the electrocution process, allowing for precise control over when and how the high-voltage net is activated to repel mosquitoes.

5. Voltage Multiplier:

The voltage multiplier is employed to further increase the voltage to levels suitable for mosquito repellency. This component is essential for creating a potent electric field that is effective in deterring mosquitoes. The voltage multiplier enhances the overall performance of the system, ensuring that the net can effectively repel mosquitoes attempting to pass through.

In summary, the theoretical foundation of the project is built upon the synergy of these key components – the 555 timer for pulse generation, the bridge rectifier for converting AC to DC, the transformer for voltage adjustment, the MOSFET switch for precise control, and the voltage multiplier for boosting the repellent capabilities. Together, these components form a robust electric mosquito repellent net that addresses real-world problems, providing a solution that can be easily replicated and implemented in various settings, including homes, schools, hospitals, and dorm rooms.

Required components:

1.Battery.
2.Capacitor(0.01 uF ceramic and electrolyte, 0.1nF)
3.Diode.(1N4007)
4. 555 timer IC.
5.Piezo buzzer.
6.SPST switch.
7.Resistor (760,1.5K,100,2.2k,22nF 2KV)
8. Variable resistor.
9.LED
10. Transformer.
11.Transistor.
12.On off switch.
13. Bread board.
14.Vero board.
15.Jumper wires.

Methodology:

The methodology of this project can be broken down into mainly the two following processes

1. Use of High Frequency

Sound of any frequency above 20 kHz is termed as ultrasonic sound. Animals like cats, dogs, insects, and mosquitoes are able to hear this ultrasonic sound. In mosquitoes, this feature is attributed to the presence of sensory structures in their antenna.Usually ultrasound is transmitted by male mosquitoes and received by female mosquitoes. However after breeding, female mosquitoes generally avoid the ultrasound and we can use this to repel mosquitos.The ultrasound produces a stress on the antennae of the mosquitoes and repels them away.In simple words, a simple circuit is designed which can produce ultrasound in the frequency range of 20 kHz to 38 kHz, which can scare away mosquitoes.

Once the switch is closed, the 555 timer gets the power supply. As per the inner circuit, initially the capacitor voltage will be zero and hence voltage at threshold and trigger pin will be zero. As the capacitor charges through resistors Ra and Rb, at a certain point voltage at threshold pin is less than the capacitor voltage. This causes a change in timer output. The capacitor now starts discharging through resistor Rb, i.e. the discharge pin and continues so until the output voltage is back to the original. Thus the output signal is an oscillating signal with frequency 38 KHz. The output from this astable multivibrator circuit drives a 38 KHz piezo buzzer, producing ultrasound at regular repetitions. On varying the value of potentiometer, the output frequency can also be varied.

Frequency of output signal produced by a 555 astable multivibrator is given by:

$$F = 1.44((Ra+Rb*2)*C)$$

Here Ra is the value of resistor between pin 7 and Vcc, Rb is value of resistor between pins 7 and 6 and C is value of capacitor between pin 6 and ground.

Let C = 0.01 microFarad

F = 38 kHz

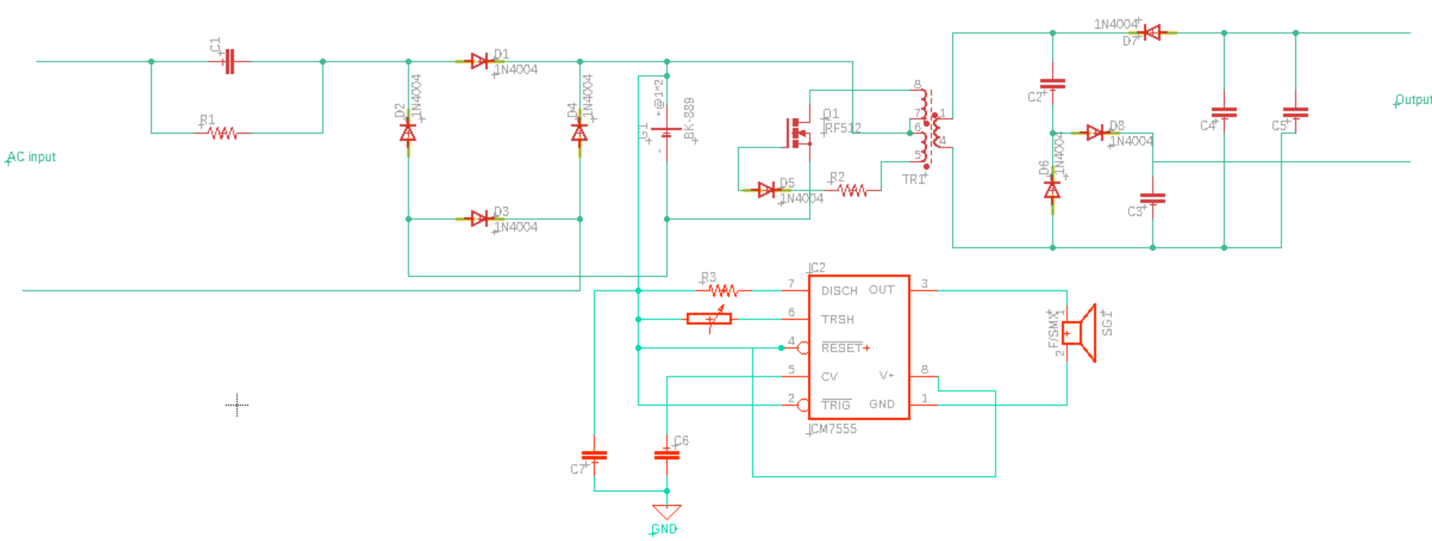
2. Use of High Voltage

We can zap the mosquitos which have somehow passed the high frequency barrier with high voltage that can kill mosquitoes but will not hurt humans as it is not that high amount of voltage.

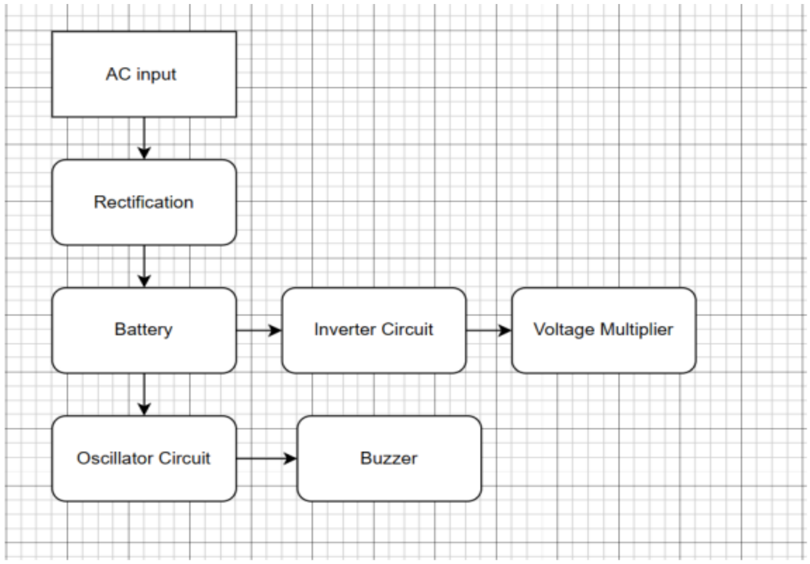
It will get power from a rechargeable battery. So it will have a charging circuit.Here we will give input of 220V AC and using diodes we will make a **rectifier** circuit to charge a DC battery.We will make a bridge rectifier. This circuit converts the low DC voltage to high voltage AC voltage about 200 to 230 V.The Blocking oscillation concept is used in this **inverter** circuit, wherein only a single NPN transistor and a center-tapped transformer is used.Here is a small EE-type ferrite core transformer used and no air gaps between them.

As we see, a high voltage generator circuit converts the voltage about 200 to 230 V, which could not build up the required voltage to create the spark between two electrodes in the mosquito killer bat. Therefore a **voltage multiplier** circuit needed to step up the voltage.voltage multiplier circuit is built using diodes and capacitors

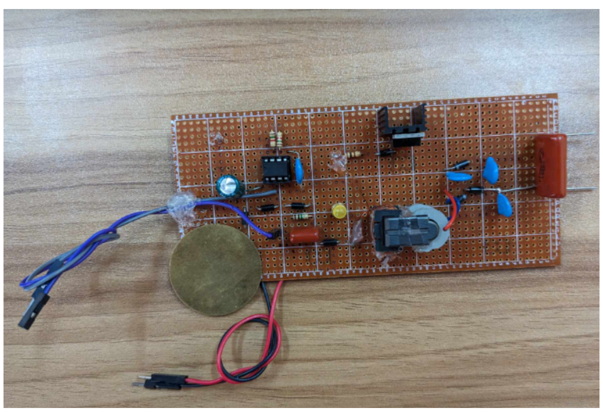
Circuit Diagram:

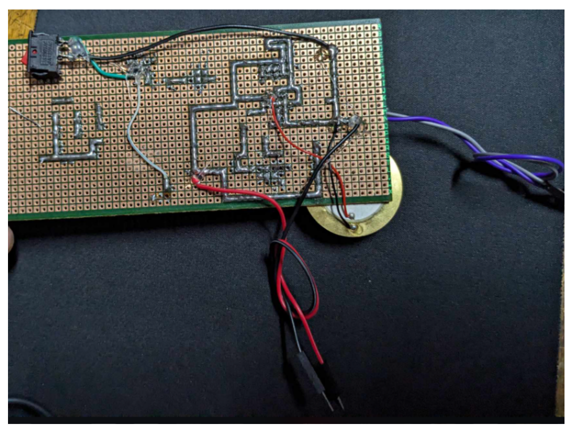


Block diagram:



Project Output:





Front View **Back View**

Upon completing the circuit of our zapper, this was the final result. While working on this, we mainly struggled in two areas -

i) soldering: this was a very tedious and time consuming process to solder the components on the veroboard.

ii) limited space: fitting the components in a tiny area was also a challenge.

We carried out tests to check whether the circuit was working correctly or not. The results of those tests were satisfactory.

The most important test was after the completion of the circuit , we had to see if the output was discharging properly or not. To check this, we brought terminals of the big red capacitor in contact with a metal, and then there was a ***spark*** as a jolt of electricity passed through the metal, and thus we concluded that our circuit was working perfectly.

Analysis:

Our mosquito zapper may be mistaken for a taser gun, because it works in a similar fashion. But the output voltage of taser guns goes up to around 50000V, whereas our zapper is giving an output voltage 3000V, which is enough to kill mosquitos.

A rectifier and step up transformer is used in this zapper circuit. The idea is that an ac input will be given and then that will be converted into dc with the help of the bridge rectifier (it will be around 3.7V DC), and then that DC voltage would be increased to around 1000V by using a step-up transformer. Then the voltage will be again multiplied with voltage multiplier circuit.And the output will be around 2500-3000V.

Whenever working with high voltage is concerned, safety is of paramount importance, so keeping that in mind, all the work was carried out by maintaining a safety margin. We were very careful not to overcharge the capacitors, which sometimes might cause them to burst, and we were also very careful while providing the other necessary connections in order to avoid any accidents.

Cost Analysis:
A detailed breakdown of the cost for each of the components utilized in our project is given below.

Components List	Cost(taka)
battery	60
diode	5*10=50
capacitor	5*9=45
555 timer	15
Piezo buzzer	20
SPST switch	70
resistor	100
Variable resistor	50
LED	20
transformer	380
transistor	25
ON-OFF switch	10
Bread board	50
Vero board	70

So, the total cost in our project is around **965 taka**.

Discussion:

Our power electronics course taught us about various components and their applications in real life situations. We used this knowledge creatively to design a Mosquito Repellent system for our project. Our system has two features: one that uses high voltage and another that uses high frequency (38KHz). This project was a valuable learning experience for our team and we believe that it will benefit us in our future projects.